

OCTOBER 2026

Predictive Mock Test SOLUTION BOOK

PHYSICS + CHEMISTRY PATHWAY

81

questions

3

modules

40

minutes per module

MODULES IN THIS BOOK

PDF PAGE

01	Mathematics 1	2
02	Physics	30
03	Chemistry	58

ThrivingScholars

Original practice material for ESAT preparation.

Independent Thriving Scholars resource.

Not an official or actual examination paper.

MODULE 01

Mathematics 1

QUESTIONS & WORKED SOLUTIONS

Physics + Chemistry pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

A rectangular card has a length recorded as 12 cm and a width recorded as 8 cm. Both measurements are rounded to the nearest centimetre. What is the least upper bound for the ratio of its actual length to its actual width?

A $\frac{3}{2}$

B $\frac{5}{3}$

C $\frac{23}{15}$

D $\frac{25}{17}$

E $\frac{25}{16}$

Worked solution

Answer B

Write the actual length and width as L and W . Their bounds are $11.5 \leq L < 12.5$ and $7.5 \leq W < 8.5$. The largest ratios use a length approaching 12.5 and the smallest width, 7.5. Thus $\frac{L}{W} < \frac{12.5}{7.5} = \frac{5}{3}$, and ratios can approach $\frac{5}{3}$ as closely as desired. The least upper bound is $\frac{5}{3}$.

Trap: using both upper bounds. To make a positive ratio as large as possible, make its numerator large and its denominator small. The upper bound need not be attained.

Question 02

For a real number k , the equation $x^2 - (k + 4)x + 2k = 0$ has one root strictly between 0 and 1, and its other root is greater than 2. What is the set of all possible values of k ?

A $k < 0$

B $0 \leq k \leq 3$

C $k > 3$

D $0 < k < 3$

E $k < 0$ or $k > 3$

Worked solution

Answer D

Let $p(x) = x^2 - (k + 4)x + 2k$. A root in $(0, 1)$ requires $p(0) > 0$ and $p(1) < 0$, giving $k > 0$ and $k < 3$. Also $p(2) = -4 < 0$, so after the first crossing the other root is beyond 2.

Trap: checking only the discriminant and forgetting where the roots lie.

Question 03

What is the value of x that satisfies $\frac{9^{x+1} 27^{2-x}}{3^{x-1}} = 81$?

A $\frac{5}{2}$

B $\frac{3}{2}$

C 2

D $\frac{1}{2}$

E $\frac{7}{2}$

Worked solution

Answer A

Write every number as a power of 3. The equation becomes $3^{2(x+1)+3(2-x)-(x-1)} = 3^4$.
Hence $9 - 2x = 4$, giving $x = \frac{5}{2}$.

Trap: subtract the whole denominator exponent $x - 1$, so its contribution is $-x + 1$.

Question 04

The least positive integer n leaves remainder 2 when divided by 5, and remainder 3 when divided by 7. What is the remainder when $n^2 + 4n + 1$ is divided by 35?

A 3

B 5

C 18

D 13

E 8

Worked solution

Answer E

The candidates leaving remainder 2 on division by 5 are 2, 7, 12, 17, The first that leaves remainder 3 on division by 7 is $n = 17$. Then $n^2 + 4n + 1 = 358 = 10 \cdot 35 + 8$, so the remainder is 8.

Question 05

A container holds 4 red, 3 blue and 2 green counters. Three counters are drawn without replacement. Given that exactly one of the three is green, what is the probability that the first counter drawn is red?

A $\frac{1}{7}$

B $\frac{8}{21}$

C $\frac{2}{9}$

D $\frac{4}{9}$

E $\frac{1}{2}$

Worked solution

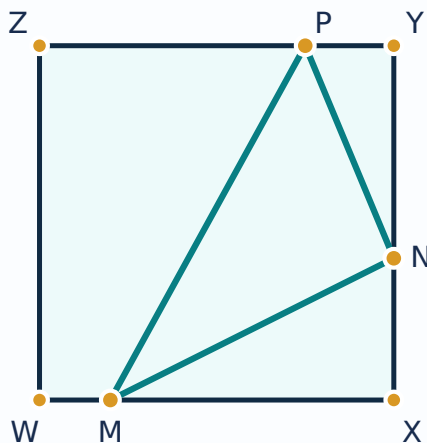
Answer B

Given that exactly one of the three counters is green, the green counter is equally likely to occupy any of the three positions. Therefore the first counter is non-green with probability $\frac{2}{3}$. Conditional on its being non-green, each of the seven non-green counters is equally likely to be first, and four are red. Hence the required probability is $\frac{2}{3} \times \frac{4}{7} = \frac{8}{21}$.

Trap: conditioning changes the sample space; the unconditional probability $\frac{4}{9}$ is not the answer.

Question 06

$WXYZ$ is a square of side 1. Points M, N, P lie on WX, XY, YZ , respectively, with $WM : MX = 1 : 4$, $XN : NY = 2 : 3$, and $YP : PZ = 1 : 3$. What is the area of triangle MNP ?



Schematic diagram; not to scale.

A $\frac{1}{5}$

B $\frac{1}{4}$

C $\frac{29}{100}$

D $\frac{3}{10}$

E $\frac{7}{20}$

Worked solution

Answer C

The ratios give $WM = \frac{1}{5}$, $MX = \frac{4}{5}$, $XN = \frac{2}{5}$, $NY = \frac{3}{5}$, $YP = \frac{1}{4}$, and $PZ = \frac{3}{4}$. Subtract the two right triangles MXN , NYP , and the trapezium $WMPZ$ from the unit square.

The two triangle areas are $\frac{1}{2} \cdot \frac{4}{5} \cdot \frac{2}{5} = \frac{4}{25}$ and $\frac{1}{2} \cdot \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{40}$. The trapezium has parallel sides WM and ZP , separated by distance 1, so its area is $\frac{1}{2}(\frac{1}{5} + \frac{3}{4}) = \frac{19}{40}$. Thus the required area is $1 - \frac{4}{25} - \frac{3}{40} - \frac{19}{40} = \frac{29}{100}$.

Tip: subtract familiar shapes from the square; no general coordinate-area formula is needed.

Question 07

Pumps A and B fill a tank at constant rates. In a trial, increasing A's rate by 20% and decreasing B's rate by 10% leaves their combined rate unchanged.

At its original rate, A alone fills the tank in 30 minutes. The empty tank is now filled using both pumps at their original rates. After 4 minutes, B is switched off and A completes the filling.

What is the total time taken to fill the tank?

A 14 minutes

B 18 minutes

C 20 minutes

D 22 minutes

E 24 minutes

Worked solution

Answer D

Let the original filling rates be a and b tanks per minute. The trial gives

$$1.2a + 0.9b = a + b \Rightarrow 0.2a = 0.1b \Rightarrow b = 2a.$$

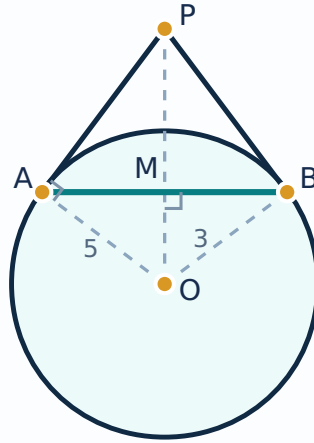
Since $a = 1/30$, the combined rate is $3a = 1/10$ tank per minute. In the first 4 minutes, the pumps fill $4/10 = 2/5$ of the tank.

A fills the remaining $3/5$ in $(3/5) \times 30 = 18$ minutes. The total is $4 + 18 = 22$ minutes.

Useful approach: Use the unchanged combined rate to find the ratio of the two original rates before calculating any times.

Question 08

A circle has centre O and radius 5. Chord AB is 3 units from O , measured perpendicularly. The tangents at A and B meet at P . What is OP ?



Schematic diagram; M is the midpoint of AB .

A $\frac{25}{3}$

B $\frac{25}{6}$

C 5

D 10

E 15

Worked solution

Answer A

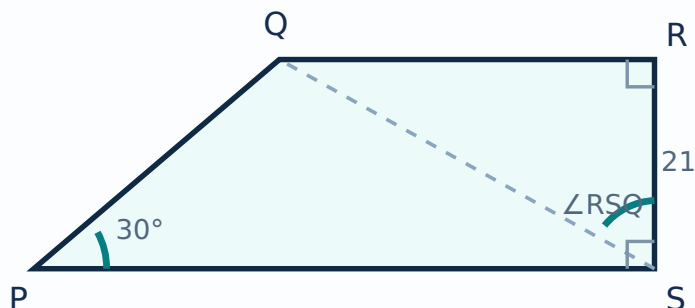
Let M be the midpoint of chord AB . The perpendicular from O to AB passes through M , and O, M, P are collinear by symmetry. We have $OM = 3$ and $OA = 5$.

Triangle OMA is right-angled at M , while triangle OAP is right-angled at A , because a radius is perpendicular to a tangent. They share the angle at O , so they are similar.

Hence $\frac{OM}{OA} = \frac{OA}{OP}$, giving $\frac{3}{5} = \frac{5}{OP}$ and $OP = \frac{25}{3}$.

Question 09

$PQRS$ is a right trapezium with $PS \parallel QR$, right angles at R, S , $RS = 21$, $\angle QPS = 30^\circ$, and $\tan \angle RSQ = \frac{4}{3}$. What is PS ?



Schematic diagram; diagonal SQ is shown for the given angle.

A $28 + 21\sqrt{3}$

B $21 + 21\sqrt{3}$

C $21 + 28\sqrt{3}$

D $49 + \sqrt{3}$

E $28 + 7\sqrt{3}$

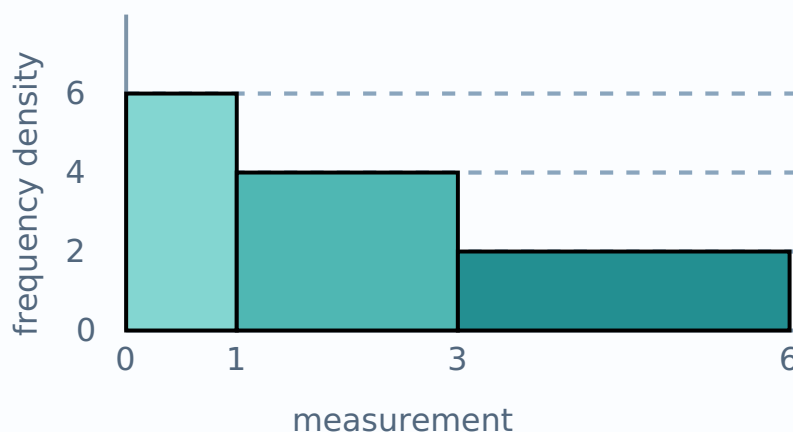
Worked solution

Answer A

In right triangle QRS , $QR/RS = 4/3$, so $QR = 28$. The horizontal distance from P to the vertical through Q is $21/\tan 30^\circ = 21\sqrt{3}$. Therefore $PS = 28 + 21\sqrt{3}$.

Question 10

A histogram represents exactly 20 measurements. Its class intervals are $0 \leq x < 1$, $1 \leq x < 3$, and $3 \leq x \leq 6$, where x is the measurement. The frequency densities are shown below.



The unequal widths are part of the histogram.

The mean of all 20 measurements is exactly 1.8. What is the greatest possible number of measurements that are at least 5?

A 3

B 5

C 8

D 4

E 6

Worked solution

Answer B

Frequency equals class width multiplied by frequency density, so the three frequencies are $1 \times 6 = 6$, $2 \times 4 = 8$, and $3 \times 2 = 6$. The total of all measurements is $20(1.8) = 36$.

Only the six observations in $[3, 6]$ can be at least 5. If all six were, the total would be at least $6(5) + 8(1) + 6(0) = 38$, which is impossible. Five are achievable: take five values 5, one value 3, eight values 1, and six values 0. These respect the stated class boundaries and sum to 36. The maximum number is therefore 5.

Question 11

Two fair six-sided dice are rolled. Given that at least one die shows 4, what is the probability that the sum of the scores is at least 9?

A $\frac{1}{3}$

B $\frac{2}{5}$

C $\frac{4}{11}$

D $\frac{5}{12}$

E $\frac{1}{2}$

Worked solution

Answer C

Given that at least one die shows 4, there are $6 + 6 - 1 = 11$ equally likely ordered outcomes. The outcomes with sum at least 9 are (4, 5), (4, 6), (5, 4), (6, 4). Therefore the probability is $4/11$.

Question 12

For positive integers n , define $a_n = n^2 + 3n + 2$. How many of the first 120 terms are divisible by 12?

A 32

B 36

C 44

D 40

E 48

Worked solution

Answer D

Factor $a_n = (n+1)(n+2)$. Divisibility by 12 requires divisibility by both 3 and 4.

For the factor 4, the even member of the two consecutive integers must be a multiple of 4. Thus n leaves remainder 2 or 3 when divided by 4.

For the factor 3, n must leave remainder 1 or 2 when divided by 3. In each block of 12 consecutive values of n , both requirements hold at remainders 2, 7, 10, 11.

There are $120/12 = 10$ complete blocks, each with 4 successes, giving 40 terms.

Question 13

The graph of $y = g(x)$ is made from straight-line segments joining, in order, the points $(-4, 3)$, $(-1, -3)$, $(2, 1)$, $(5, -2)$, $(7, 2)$. What is the sum of the x -coordinates of all its intersections with the x -axis?

A $\frac{29}{4}$

B $\frac{31}{4}$

C 8

D $\frac{33}{4}$

E $\frac{35}{4}$

Worked solution

Answer B

On the first segment, the heights 3 and -3 put the crossing halfway between $x = -4$ and $x = -1$, at $-\frac{5}{2}$. On the second segment, the height rises from -3 to 1, so zero is reached three quarters of the way from -1 to 2: $-1 + \frac{3}{4}(3) = \frac{5}{4}$.

On the third segment, the height falls from 1 to -2 , so the crossing is one third of the way from 2 to 5, at 3. The final segment crosses halfway from 5 to 7, at 6. Their sum is $-\frac{5}{2} + \frac{5}{4} + 3 + 6 = \frac{31}{4}$.

Question 14

For a real parameter a , consider $x^2 - 2(a+1)x + a^2 - 1 = 0$. For which values of a does this equation have one positive root and one negative root?

A $a \leq -1$

B $-1 \leq a < 0$

C $-1 < a < 1$

D $0 < a \leq 1$

E $a < -1$ or $a > 1$

Worked solution

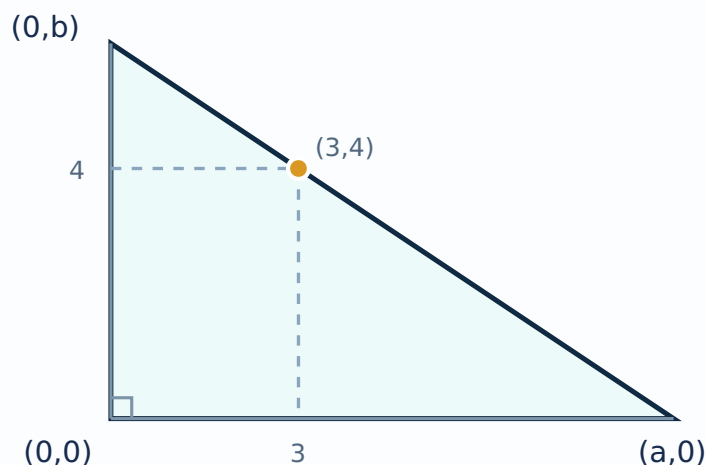
Answer C

Writing a monic quadratic as $(x-r)(x-s)$ shows that its constant term is the product rs . Here the product is $a^2 - 1$. Opposite signs require a negative product, so $a^2 - 1 < 0$, giving $-1 < a < 1$.

This condition is also sufficient: the upward-opening quadratic is negative at $x = 0$ and positive for sufficiently large positive or negative x , so it crosses the axis once on each side of zero. The endpoints $a = \pm 1$ are excluded because one root is then zero.

Question 15

A right triangle has vertices $(0,0)$, $(a,0)$, $(0,b)$, where $a, b > 0$. The point $(3,4)$ lies on its hypotenuse. What is the minimum possible area of the triangle?



Schematic diagram; not to scale.

A 18

B 20

C 32

D 28

E 24

Worked solution

Answer E

The line through $(a,0)$ and $(0,b)$ has equation $y = b - \frac{b}{a}x$, equivalently $\frac{x}{a} + \frac{y}{b} = 1$. Since $(3,4)$ lies on it, $\frac{3}{a} + \frac{4}{b} = 1$. Set $u = \frac{3}{a}$ and $v = \frac{4}{b}$, so $u + v = 1$ and $ab = \frac{12}{uv}$.

Because $(u - v)^2 \geq 0$, we have $4uv \leq (u + v)^2 = 1$. Therefore $ab \geq 48$, and the area is at least $\frac{1}{2}(48) = 24$. Equality occurs at $u = v = \frac{1}{2}$, giving $a = 6$ and $b = 8$, so this minimum is attained.

Question 16

A sequence is defined by $a_1 = 1$ and $a_{n+1} = 3a_n + 2$. What is the remainder when a_{2026} is divided by 7?

A 4

B 2

C 1

D 5

E 6

Worked solution

Answer A

The successive remainders on division by 7 are 1, 5, 3, 4, 0, 2, then repeat. Since $2026 = 6 \cdot 337 + 4$, the required remainder is the fourth one in the cycle, namely 4.

Question 17

Points $A = (-4, 7)$, $B = (5, -2)$ and $C = (3, 6)$ are given. Point P lies on AB such that $AP : PB = 2 : 1$. Point Q is the reflection of C in P , meaning a rotation of 180° about P . What is $2x_Q - y_Q$?

A -6

B -2

C 2

D 6

E 10

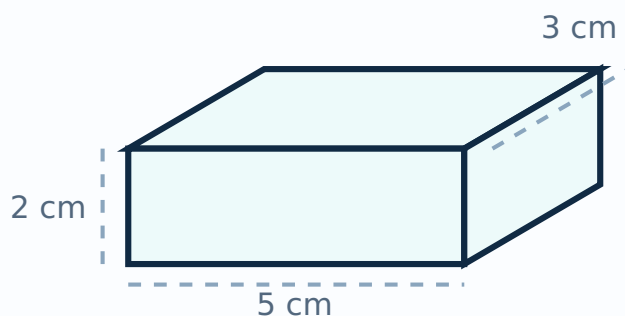
Worked solution

Answer D

$P = A + \frac{2}{3}(B - A) = (-4, 7) + \frac{2}{3}(9, -9) = (2, 1)$. Since P is the midpoint of CQ , $Q = 2P - C = (1, -4)$. Thus $2x_Q - y_Q = 2 + 4 = 6$.

Question 18

A rectangular metal block measures $2\text{ cm} \times 3\text{ cm} \times 5\text{ cm}$ and has mass 270 g . What is its density in kg m^{-3} ?



Schematic block; dimensions are not drawn to scale.

A 90

B 900

C 9000

D 27000

E 90000

Worked solution

Answer C

The block has volume $2 \times 3 \times 5 = 30\text{ cm}^3 = 3 \times 10^{-5}\text{ m}^3$. Its mass is $270\text{ g} = 0.270\text{ kg}$.
Therefore its density is $\frac{0.270}{3 \times 10^{-5}} = 9000\text{ kg m}^{-3}$.

Question 19

Let O, A, B, C be the vertices of a parallelogram, with O as the origin and $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ as adjacent sides. Thus C is the fourth vertex. Point P lies on AC with $AP : PC = 1 : 2$, and Q is the midpoint of OB . Which expression is $\overrightarrow{PQ}$?

A $\mathbf{a} + \frac{1}{6}\mathbf{b}$

B $-\mathbf{a} + \frac{1}{6}\mathbf{b}$

C $-\mathbf{a} - \frac{1}{6}\mathbf{b}$

D $\mathbf{a} - \frac{1}{6}\mathbf{b}$

E $-\mathbf{a} + \frac{1}{3}\mathbf{b}$

Worked solution

Answer B

Because OA and OB are adjacent sides, $\overrightarrow{OC} = \mathbf{a} + \mathbf{b}$, so $P = \mathbf{a} + \frac{1}{3}\mathbf{b}$. Also $Q = \frac{1}{2}\mathbf{b}$.
Therefore $\overrightarrow{PQ} = Q - P = -\mathbf{a} + \frac{1}{6}\mathbf{b}$.

Question 20

The line $y = 2x - 1$ is tangent to the graph $y = x^2 - 2kx + k + 3$, where k is real. What is the sum of the two possible values of k ?

A -3

B -2

C 3

D 1

E -1

Worked solution

Answer E

At an intersection, $x^2 - 2(k+1)x + k + 4 = 0$. Completing the square gives $(x - k - 1)^2 = k^2 + k - 3$. A tangent has exactly one point of contact, so the right-hand side must be zero.

Thus $k^2 + k - 3 = 0$, giving $k = \frac{-1+\sqrt{13}}{2}$ or $k = \frac{-1-\sqrt{13}}{2}$. The sum is -1 .

Tip: a repeated root can be recognised by completing the square; differentiation is not required.

Question 21

A quantity P is directly proportional to x^2 and inversely proportional to y . If x is increased by 20% and y is decreased by 20%, by what factor does P change?

A $\frac{4}{5}$

B $\frac{6}{5}$

C $\frac{9}{5}$

D 2

E $\frac{12}{5}$

Worked solution

Answer C

Write $P = c\frac{x^2}{y}$ for some constant c . After the changes, x is multiplied by $6/5$ and y by $4/5$, so P is multiplied by $\frac{(6/5)^2}{4/5} = \frac{9}{5}$.

Question 22

Six cards are numbered 1, 2, 3, 4, 5, 6. Two different cards are selected, with every pair equally likely.

Given that the product of their numbers is even, what is the probability that their sum is greater than 7?

A $\frac{1}{3}$

B $\frac{1}{5}$

C $\frac{5}{12}$

D $\frac{1}{2}$

E $\frac{3}{5}$

Worked solution

Answer C

There are $6 \times 5/2 = 15$ unordered pairs. The product is odd only when both cards are odd. The three odd cards form 3 pairs, so $15 - 3 = 12$ pairs satisfy the condition.

The eligible pairs whose sum exceeds 7 are

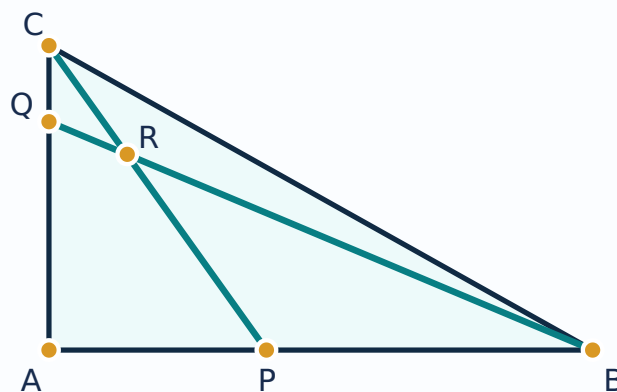
$$(2, 6), (3, 6), (4, 5), (4, 6), (5, 6).$$

The pair (3, 5) has a large enough sum but has an odd product. The required probability is $5/12$.

Useful approach: Apply the given condition to both the denominator and the favourable outcomes.

Question 23

In triangle ABC , let $A = (0, 0)$, $B = (10, 0)$, and $C = (0, 8)$. Point P lies on AB with $AP : PB = 2 : 3$, and point Q lies on AC with $AQ : QC = 3 : 1$. Lines CP and BQ meet at R . What is the x -coordinate of R ?



Schematic diagram; not to scale.

A 1

B $\frac{4}{3}$

C $\frac{5}{3}$

D $\frac{10}{7}$

E 2

Worked solution

Answer D

$P = (4, 0)$, $Q = (0, 6)$. The two lines are $CP : y = 8 - 2x$ and $BQ : y = 6 - \frac{3}{5}x$. Equating gives $2 = \frac{7}{5}x$, hence $x = 10/7$.

Question 24

Box A is chosen with probability $\frac{2}{3}$, and box B with probability $\frac{1}{3}$. Box A contains 2 red and 3 blue counters; box B contains 4 red and 1 blue counter. Two counters are drawn without replacement from the chosen box. Given that the two counters have different colours, what is the probability that box A was chosen?

A $\frac{1}{2}$

B $\frac{2}{3}$

C $\frac{7}{10}$

D $\frac{3}{4}$

E $\frac{4}{5}$

Worked solution

Answer D

Let D mean that the two drawn counters have different colours. From box A , the red-blue and blue-red branches give $P(D | A) = 2\left(\frac{2}{5}\right)\left(\frac{3}{4}\right) = \frac{3}{5}$. For B , $P(D | B) = 2\left(\frac{4}{5}\right)\left(\frac{1}{4}\right) = \frac{2}{5}$.

The probability of choosing A and then drawing different colours is $\frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$. The probability of different colours overall is $\frac{2}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{8}{15}$. Of these different-colour outcomes, the fraction originating from A is $\frac{2/5}{8/15} = \frac{3}{4}$.

Question 25

How many ordered pairs of positive integers (a, b) satisfy $ab = 360$, have highest common factor 1, and have $a + b$ a multiple of 7?

A 2

B 0

C 6

D 8

E 4

Worked solution

Answer E

Since $360 = 2^3 \cdot 3^2 \cdot 5$ and a, b are coprime, the prime powers 8, 9, 5 must be assigned wholly to one factor. The possible unordered splits are $1 + 360, 8 + 45, 9 + 40, 72 + 5$; only $9 + 40 = 49$ and $72 + 5 = 77$ are multiples of 7. Both orders count, giving 4.

Question 26

Data set A has mean 14, and data set B has mean 20. The mean of all observations in the two data sets combined is 16. An observation of value 10 from A is exchanged with an observation of value 34 from B . The two data sets then have the same mean. How many observations are there altogether?

A 6

B 12

C 18

D 24

E 36

Worked solution

Answer C

Exchanging the two observations changes neither the combined total nor the size of either data set. The combined mean remains 16, so the two equal new means must both be 16.

The total in A increases by $34 - 10 = 24$, while its mean increases by $16 - 14 = 2$. Therefore A contains $24/2 = 12$ observations. The total in B decreases by 24, while its mean decreases by $20 - 16 = 4$, so B contains $24/4 = 6$ observations. Altogether there are $12 + 6 = 18$.

Trap: averaging the two original means without accounting for the different data set sizes.

Question 27

Two geometrically similar solid objects are made from the same uniform material. Their total surface areas are in the ratio $4 : 9$, smaller to larger. The mass of the larger object is 95 g greater than the mass of the smaller object. What is the mass of the smaller object?

A 40 g

B 20 g

C 76 g

D 95 g

E 190 g

Worked solution

Answer A

The surface area ratio $4 : 9$ gives a length ratio $2 : 3$. The volume ratio is therefore $2^3 : 3^3 = 8 : 27$. Because the objects are made from the same uniform material, their masses also have ratio $8 : 27$.

The difference of $27 - 8 = 19$ ratio parts is 95 g, so one part is 5 g. The smaller mass is $8 \times 5 = 40$ g.

Trap: applying the surface area ratio directly to mass gives 76 g; using the length ratio gives 190 g.

MODULE 02

Physics

QUESTIONS & WORKED SOLUTIONS

Physics + Chemistry pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

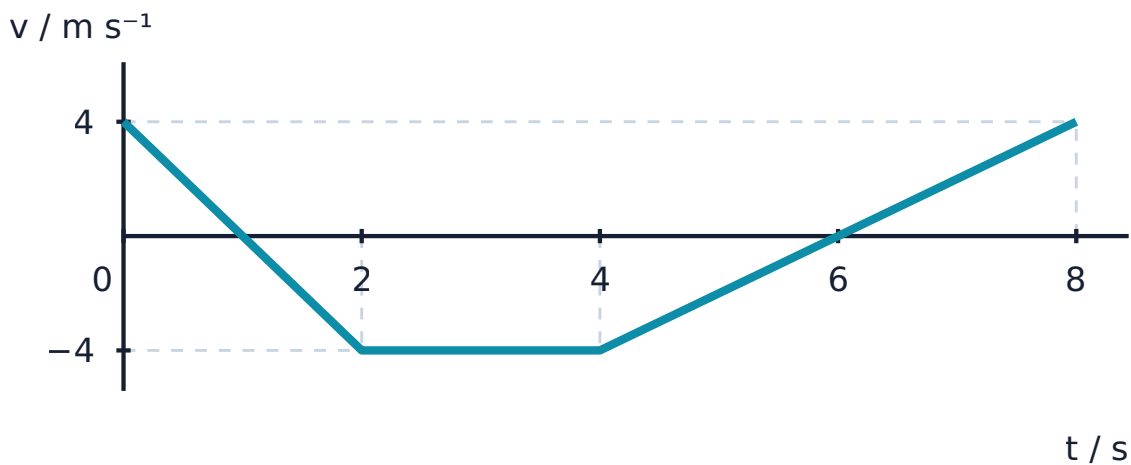
Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

The velocity of a particle varies with time as shown. What is the ratio of the total distance travelled to the magnitude of the displacement during the 8.0 s interval?



A 2

B $5/2$

C 3

D $7/2$

E 4

Worked solution

Answer B

From 0–2 s the signed area is zero, but the two triangular parts contribute 4 m of distance. From 2–4 s the particle has velocity -4 m s^{-1} , giving -8 m of displacement and 8 m of distance. From 4–8 s the signed area is again zero, while the two triangles contribute 8 m of distance. Thus the displacement is -8 m , its magnitude is 8 m, and the total distance is $4+8+8=20 \text{ m}$. The ratio is $20/8=5/2$.

Useful approach: Compute signed area and absolute area separately; a zero net area can still contain substantial distance.

Question 02

A solid cylinder A has radius r , height h and mass m . A second cylinder B is made from the same material, with radius $2r$ and height $h/2$. Both cylinders stand on circular bases. Cylinder B rests on a horizontal floor and A is placed centrally on B. What is the ratio of the pressure exerted on the floor by the stack to the pressure exerted by A alone on the floor?

A 1

B $1/4$

C $3/2$

D $1/2$

E $3/4$

Worked solution

Answer E

The volume of B is $\pi(2r)^2(h/2) = 2\pi r^2 h$, twice the volume of A, so B has mass $2m$. The stack therefore weighs $3mg$. The floor contact area is B's base area, $4\pi r^2$, so the stack pressure is $3mg/(4\pi r^2)$. A alone would exert $mg/(\pi r^2)$. Their ratio is $3/4$.

Useful approach: Find both the new weight and the new floor contact area before taking the ratio.

Question 03

An ideal battery supplies a network consisting of one resistor R in series with two resistors R in parallel. This network is replaced by two branches in parallel, each branch containing two resistors R in series. The battery voltage is unchanged. What is the ratio of the new total power to the original total power?

A $1/2$

B $2/3$

C 1

D $3/2$

E 2

Worked solution

Answer D

Initially, $R_{\text{eq}} = R + (R/2) = 3R/2$. In the replacement network each branch is $2R$, so the two branches have equivalent resistance R . At fixed voltage, total power is proportional to $1/R_{\text{eq}}$. Hence $P_{\text{new}}/P_{\text{old}} = (1/R)/(2/(3R)) = 3/2$.

Useful approach: Reduce the network before using $P = V^2/R$; the battery voltage is the fixed quantity.

Question 04

A spring stores 0.18 J when its extension is 3.0 cm. It obeys Hooke's law for all extensions up to 5.0 cm. What is the average magnitude of the spring force while it is extended at constant speed from 3.0 cm to 5.0 cm?

A 8 N

B 12 N

C 16 N

D 20 N

E 32 N

Worked solution

Answer C

At 3.0 cm, $0.18 = \frac{1}{2}kx^2$, so $k = 2 \times 0.18 / 0.030^2 = 400 \text{ N m}^{-1}$. The spring energies at 3.0 cm and 5.0 cm are therefore 0.18 J and $\frac{1}{2} \times 400 \times 0.050^2 = 0.50 \text{ J}$. The work done in the interval is 0.32 J over 0.020 m, so the average force is $0.32 / 0.020 = 16 \text{ N}$. Equivalently, it is the mean of the endpoint forces 12 N and 20 N.

Useful approach: For a linear force-extension relation, average force over an interval is the mean of the endpoint forces.

Question 05

A detector has a constant background count of 12 counts per minute. The total count from a radioactive source is 172 counts per minute initially and 92 counts per minute 20 minutes later. How many minutes after the initial reading will the total recorded count rate be 32 counts per minute? Treat the readings as mean count rates.

A 20 min

B 30 min

C 40 min

D 60 min

E 80 min

Worked solution

Answer D

Subtracting the background gives source counts of 160 initially and 80 after 20 minutes, so the half-life is 20 minutes. A total count of 32 means a source count of $32 - 12 = 20$. This is $160 \rightarrow 80 \rightarrow 40 \rightarrow 20$, three half-lives, so the time is $3 \times 20 = 60$ min.

Useful approach: Translate the requested total count back into a source count before counting half-lives.

Question 06

A vehicle travels one quarter of the total journey distance at 12 m s^{-1} , one half at 18 m s^{-1} and the final quarter at 36 m s^{-1} . What is its average speed for the whole journey?

A 18 m s^{-1}

B 15 m s^{-1}

C 20 m s^{-1}

D 12 m s^{-1}

E 24 m s^{-1}

Worked solution

Answer A

Let the journey length be D . The total time is $D/(4 \times 12) + D/(2 \times 18) + D/(4 \times 36) = D/48 + D/36 + D/144 = 8D/144 = D/18$. Thus the average speed is $D/(D/18) = 18 \text{ m s}^{-1}$.

Useful approach: Set a convenient total distance, such as 144 m, if fractions feel awkward.

Question 07

A 0.20 kg trolley moving at 6.0 m s^{-1} collides with a stationary 0.40 kg trolley and the two stick together. What fraction of the initial kinetic energy remains immediately after the collision?

A $1/9$

B $1/3$

C $1/2$

D $2/3$

E $3/4$

Worked solution

Answer B

Momentum gives $0.20 \times 6 = 0.60v$, so $v = 2 \text{ m s}^{-1}$. Initial kinetic energy is 3.6 J; final kinetic energy is 1.2 J. The fraction remaining is $1.2/3.6 = 1/3$.

Useful approach: Use momentum during the collision, then compare kinetic energies; sticking does not conserve kinetic energy.

Question 08

Car A has mass m , speed v and is stopped by a constant braking force F . Car B has mass $2m$, speed $2v$ and is stopped by a constant braking force $2F$. What is the ratio of B's stopping distance to A's?

A $1/2$

B 1

C 2

D 4

E 8

Worked solution

Answer D

Stopping distance is proportional to kinetic energy divided by braking force. B has $\frac{1}{2}(2m)(2v)^2 = 8 \times \frac{1}{2}mv^2$, while its braking force is only twice as large. Therefore its distance is $8/2 = 4$ times A's.

Useful approach: Write the proportionality $d \propto mv^2/F$ before inserting the scale factors.

Question 09

In a wind tunnel, the resistive forces on two similarly shaped falling objects are measured.

Speed / m s^{-1}	2	4	6
Object X drag / N	1	4	9
Object Y drag / N	2	8	18

Object X weighs 9 N and object Y weighs 18 N. Ignoring upthrust, which statement is correct about their terminal velocities in still air?

- A** Both objects reach terminal velocity at 6 m s^{-1} .
- B** Y has the greater terminal speed because it is heavier.
- C** X reaches terminal velocity at 4 m s^{-1} and Y at 6 m s^{-1} .
- D** Neither object reaches terminal velocity in the range shown.
- E** X reaches terminal velocity at 6 m s^{-1} and Y at 4 m s^{-1} .

Worked solution

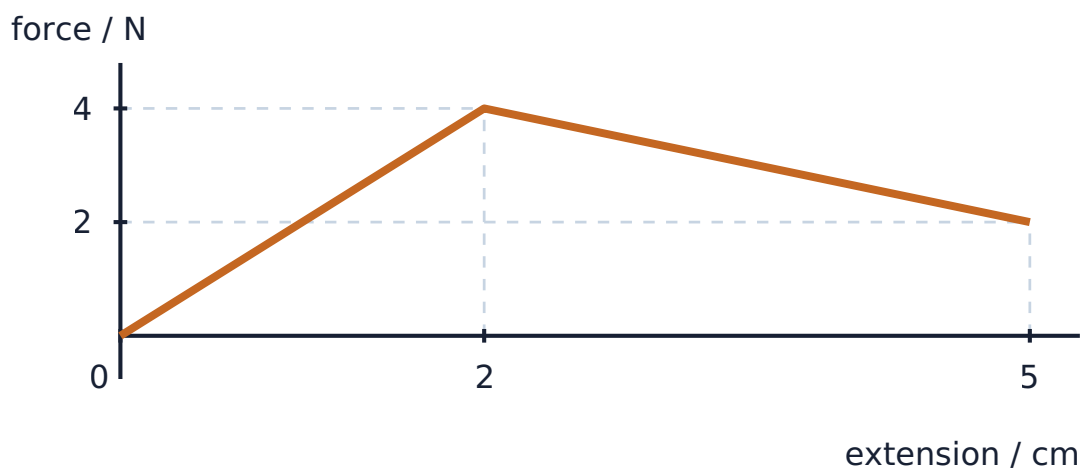
Answer A

Terminal velocity occurs when drag equals weight. At 6 m s^{-1} , X has drag 9 N, equal to its weight, and Y has drag 18 N, equal to its weight. Both therefore reach terminal velocity at 6 m s^{-1} .

Useful approach: Do not infer terminal velocity from mass alone: compare the resistive force with the weight.

Question 10

The graph shows the force applied to a wire as it is stretched slowly. What is the work done on the wire in increasing its extension from zero to 5.0 cm?



A 0.09 J

B 0.16 J

C 0.11 J

D 0.20 J

E 0.13 J

Worked solution

Answer E

The work done is the area under the loading force-extension graph. The first triangle has area $\frac{1}{2} \times 4 \times 0.020 = 0.040$ J. The second section is a trapezium with area $\frac{1}{2}(4+2) \times 0.030 = 0.090$ J. Total work done is 0.130 J.

This loading graph alone does not determine the recoverable elastic energy; that would also require information about unloading.

Useful approach: Convert centimetres to metres before multiplying force by extension.

Question 11

An NTC thermistor and a fixed resistor are connected in series across an ideal battery. A voltmeter is connected across the thermistor. The temperature of the thermistor decreases. Which change occurs?

- A** Thermistor resistance decreases, current increases, voltmeter reading increases.
- B** Thermistor resistance increases, current increases, voltmeter reading decreases.
- C** Thermistor resistance decreases, current decreases, voltmeter reading decreases.
- D** Thermistor resistance and current are unchanged, but the voltmeter reading increases.
- E** Thermistor resistance increases, current decreases, voltmeter reading increases.

Worked solution

Answer E

An NTC thermistor has greater resistance at lower temperature. The total series resistance therefore increases, so the current decreases. The thermistor's share of the fixed battery voltage increases because its resistance is now a larger fraction of the total. The voltmeter reading increases.

Useful approach: Track the chain: NTC cooler → resistance rises → total current falls → larger fraction of the fixed voltage lies across the thermistor.

Question 12

Water waves travel at 12 cm s^{-1} in two trials in a ripple tank. The water depth is unchanged. In each trial, straight wavefronts approach a gap in a barrier.

Trial	Frequency / Hz	Gap width / cm
P	4.0	6.0
Q	2.0	3.0

Compared with trial P, how do the wavelength and the spreading of the waves after the gap change in trial Q?

- A** The wavelength is twice as large; there is less spreading.
- B** The wavelength is twice as large; there is more spreading.
- C** The wavelength is half as large; there is more spreading.
- D** The wavelength is half as large; there is less spreading.
- E** The wavelength is unchanged; there is more spreading.

Worked solution

Answer B

Using $v = f\lambda$, the wavelengths are $12/4 = 3.0 \text{ cm}$ in P and $12/2 = 6.0 \text{ cm}$ in Q. The wavelength doubles.

Diffraction produces greater spreading when the wavelength is larger relative to the gap width. The wavelength-to-gap ratios are $3/6 = 1/2$ in P and $6/3 = 2$ in Q, so Q produces more spreading.

Useful approach: Compare wavelength with gap width; the absolute gap width alone does not determine diffraction.

Question 13

A fixed electrical power is transmitted through cables whose resistance does not change. The transmission voltage is increased by a factor of 4. By what factor does the power wasted in the cables change?

A It is unchanged.

B It becomes 1/2 as large.

C It becomes 1/4 as large.

D It becomes 1/8 as large.

E It becomes 1/16 as large.

Worked solution

Answer E

Since transmitted power is fixed and $P=VI$, increasing voltage fourfold makes current one quarter as large. Cable loss is I^2R , so the loss becomes $(1/4)^2 = 1/16$ as large.

Useful approach: Apply the voltage change to current first, then square the current for cable loss.

Question 14

An ideal transformer has 600 turns on its primary coil and 120 turns on its secondary coil. The secondary supplies a 24 V, 2.0 A load. What are the primary voltage and primary current?

A 24 V and 2.0 A

B 120 V and 0.40 A

C 120 V and 10 A

D 4.8 V and 10 A

E 600 V and 0.08 A

Worked solution

Answer B

The turns ratio is $N_p/N_s = 600/120 = 5$, so the primary voltage is $5 \times 24 = 120$ V. For an ideal transformer, input and output powers are equal: $V_p I_p = 24 \times 2$. Hence $I_p = 48/120 = 0.40$ A.

Useful approach: Use the turns ratio for voltage, then power conservation for current.

Question 15

10 g of ice at 0 °C is added to 100 g of water at 30 °C in an insulated container of negligible heat capacity. Use $c = 4.0 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$ and latent heat of fusion $L = 320 \text{ J g}^{-1}$. What is the final temperature?

A 0 °C

B 10 °C

C 20 °C

D 25 °C

E 30 °C

Worked solution

Answer C

Cooling the water to 0 °C releases $100 \times 4 \times 30 = 12000 \text{ J}$. Melting the ice uses $10 \times 320 = 3200 \text{ J}$, leaving 8800 J to warm 110 g of water. The temperature rise is $8800 / (110 \times 4) = 20 \text{ }^{\circ}\text{C}$, so the final temperature is 20 °C.

Useful approach: Check the phase-change stage first; only leftover energy heats the final mixture.

Question 16

A fixed amount of gas initially has pressure p , volume V and absolute temperature T . It is compressed at constant temperature to volume $V/2$. It is then heated at constant volume until its pressure is $3p$. For this gas at constant volume, pressure is directly proportional to absolute temperature. What is its final absolute temperature in terms of T ?

A $T/2$

B $2T/3$

C T

D $3T/2$

E $2T$

Worked solution

Answer D

After the isothermal compression, pressure doubles to $2p$. At constant volume, pressure is proportional to absolute temperature. Increasing pressure from $2p$ to $3p$ therefore multiplies temperature by $3/2$, giving $3T/2$.

Useful approach: Record the intermediate pressure before applying the second process.

Question 17

A light wave has wavelength λ in air. It enters a transparent material where its speed is $3c/4$, compared with speed c in air. How many complete oscillations occur while the wave travels normally through the parallel faces of a slab of thickness 6λ inside the material?

A 4

B 6

C 8

D 10

E 12

Worked solution

Answer C

Frequency remains unchanged at the boundary. Therefore the material wavelength is $3\lambda/4$. The number of wavelengths in the slab is $6\lambda/(3\lambda/4) = 8$, so eight complete oscillations occur.

Useful approach: Find the new wavelength before dividing the slab thickness by it.

Question 18

A source is tested using a detector in a fixed position. The background is 20 counts per minute. The table gives mean total count rates.

Absorber between source and detector	Total count rate / min^{-1}
None	500
Thin paper	260
Aluminium sheet	80

For this experiment, paper absorbs all alpha radiation and has negligible effect on beta and gamma radiation. The aluminium absorbs all alpha and beta radiation and has negligible effect on gamma radiation.

What fraction of the detected source count rate with no absorber is due to alpha and gamma radiation combined?

A $\frac{3}{8}$

B $\frac{1}{2}$

C $\frac{5}{8}$

D $\frac{3}{4}$

E $\frac{4}{5}$

Worked solution

Answer C

The source count rate without an absorber is $500 - 20 = 480 \text{ min}^{-1}$.

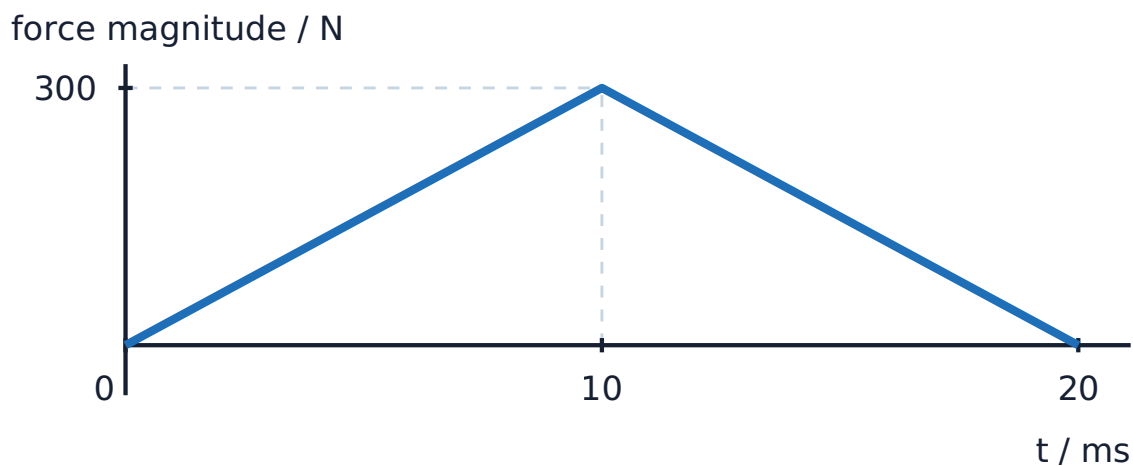
The drop caused by paper is the detected alpha contribution: $500 - 260 = 240 \text{ min}^{-1}$. The count remaining behind aluminium is gamma plus background, so the detected gamma contribution is $80 - 20 = 60 \text{ min}^{-1}$.

The required fraction is $(240 + 60)/480 = 300/480 = 5/8$. The beta contribution, $260 - 80 = 180 \text{ min}^{-1}$, accounts for the remaining $3/8$.

Useful approach: Subtract the background from a remaining total count, but not a second time from the difference between two total counts.

Question 19

A 0.20 kg ball travels to the right at 10 m s^{-1} and rebounds from a stationary wall. The force exerted by the wall on the ball is to the left and has the magnitude shown in the triangular force-time graph. Neglect any other horizontal impulse during the collision. What is the speed of the ball immediately after the collision?



A 5 m s^{-1}

B 10 m s^{-1}

C 15 m s^{-1}

D 20 m s^{-1}

E 30 m s^{-1}

Worked solution

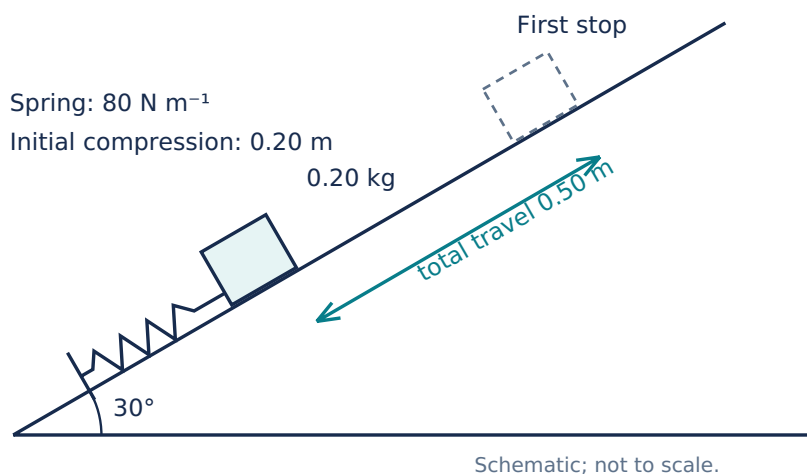
Answer A

The impulse is the triangular area $\frac{1}{2} \times 300 \times 0.020 = 3 \text{ N s}$ to the left. Taking right as positive, the initial momentum is $0.20 \times 10 = +2 \text{ N s}$ and the final momentum is $2 - 3 = -1 \text{ N s}$. The final velocity is $-1/0.20 = -5 \text{ m s}^{-1}$, so the rebound speed is **5 m s^{-1}** .

Useful approach: Convert milliseconds to seconds and assign a sign before applying $\Delta p = \text{impulse}$.

Question 20

A 0.20 kg block is initially at rest on a rough 30° slope, against a light spring of spring constant 80 N m^{-1} compressed by 0.20 m along the slope. The block is released, moves up the slope and loses contact with the spring when it reaches its natural length. It first comes to rest after travelling a total of 0.50 m along the slope, measured from its initial compressed position. The friction force has constant magnitude; neglect other energy losses. Taking $g = 10 \text{ N kg}^{-1}$, what is the friction force on the block?



A 1.1 N

B 1.6 N

C 2.2 N

D 3.2 N

E 4.4 N

Worked solution

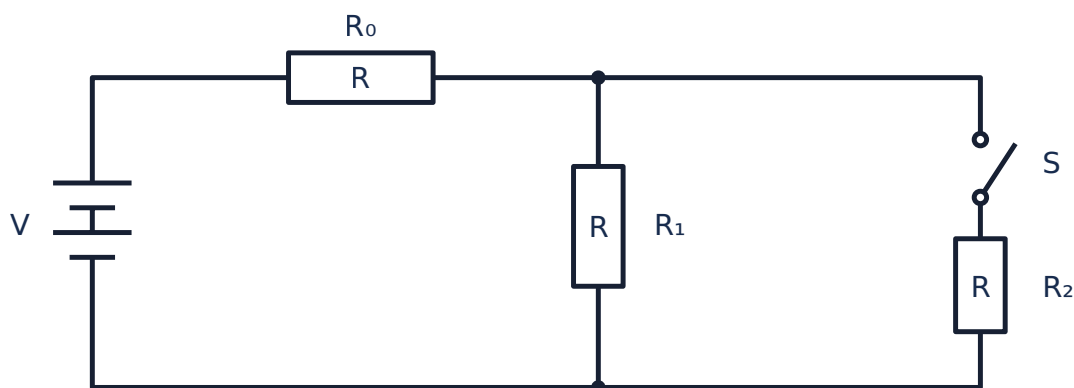
Answer C

The spring supplies $\frac{1}{2}kx^2 = \frac{1}{2} \times 80 \times 0.20^2 = 1.6 \text{ J}$. The vertical rise is $0.50 \sin 30^\circ = 0.25 \text{ m}$, so gravitational potential energy increases by $0.20 \times 10 \times 0.25 = 0.50 \text{ J}$. Friction therefore removes 1.10 J. Since its distance is 0.50 m, the friction force is $1.10/0.50 = 2.2 \text{ N}$.

Useful approach: Use spring energy = gravitational gain + work against friction.

Question 21

An ideal battery of fixed voltage V supplies the circuit shown. Each resistor has resistance R . Switch S is initially closed and is then opened. What is the ratio of the new power dissipated in R_1 to its original power?



Switch S shown open.

A $4/9$

B $3/4$

C 1

D $3/2$

E $9/4$

Worked solution

Answer E

With S closed, the parallel pair has resistance $R/2$, so the total resistance is $3R/2$. The voltage across R_1 is one third of the battery voltage: $V/3$. Its power is therefore $P_{\text{old}} = (V/3)^2/R = V^2/(9R)$.

With S open, R_0 and R_1 are in series and share the voltage equally. The voltage across R_1 becomes $V/2$, giving $P_{\text{new}} = V^2/(4R)$. Hence $P_{\text{new}}/P_{\text{old}} = 9/4$.

Useful approach: Track the voltage across the named resistor. Its power increases even though the total battery power decreases.

Question 22

A simple AC generator rotates at constant speed in a uniform magnetic field. At position A, the active sides of the coil move parallel to the field. After a quarter-turn, at position B, they move perpendicular to the field. Which statement is correct? Any change in rotation frequency leaves the coil and magnetic field unchanged.

- A** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency doubles both peak voltage and output frequency.
- B** The voltage is maximum in magnitude at A and zero at B; doubling rotation frequency doubles both peak voltage and output frequency.
- C** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency leaves peak voltage unchanged and doubles output frequency.
- D** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency doubles peak voltage and halves output frequency.
- E** The voltage is zero at A and maximum in magnitude at B; doubling rotation frequency halves peak voltage and doubles output frequency.

Worked solution

Answer A

Motion parallel to the field cuts no field lines, so the induced voltage is zero at A. Perpendicular motion gives the greatest rate of cutting, so the voltage has maximum magnitude at B. Doubling the rotation frequency doubles both the peak voltage and the output frequency.

Useful approach: Use the direction and speed of motion across the magnetic field to determine the induced voltage.

Question 23

Two slabs A and B have the same cross-sectional area and thickness. They are in perfect thermal contact, with their sides insulated. The outer face of A is held at $100\text{ }^{\circ}\text{C}$ and the outer face of B at $20\text{ }^{\circ}\text{C}$. Their thermal conductivities are k and $3k$, respectively.

For a fixed area and thickness, the rate of heat transfer through a slab is proportional to its thermal conductivity multiplied by the temperature difference across it. What is the temperature at the interface in steady state?

A $40\text{ }^{\circ}\text{C}$

B $30\text{ }^{\circ}\text{C}$

C $60\text{ }^{\circ}\text{C}$

D $80\text{ }^{\circ}\text{C}$

E $20\text{ }^{\circ}\text{C}$

Worked solution

Answer A

In steady state, the same rate of heat transfer passes through both slabs. If the interface temperature is $T\text{ }^{\circ}\text{C}$, the supplied relationship gives $k(100-T) = 3k(T-20)$. Cancelling k gives $100-T = 3T-60$, so $4T = 160$ and **$T = 40\text{ }^{\circ}\text{C}$** .

Equivalently, the less conductive slab A has three times the temperature drop of B: $60\text{ }^{\circ}\text{C}$ across A and $20\text{ }^{\circ}\text{C}$ across B.

Useful approach: The less conductive slab has the larger temperature drop.

Question 24

An ultrasound probe is placed on the front surface of a metal plate of thickness 0.30 m. Pulses travel normally to the plate faces. Measured from emission, an echo from the far surface returns after 0.10 ms and an echo from a crack returns after 0.020 ms. Assuming the ultrasound speed is the same on both paths, how far below the front surface is the crack?

A 0.03 m

B 0.06 m

C 0.10 m

D 0.12 m

E 0.24 m

Worked solution

Answer B

The far-surface echo travels 0.60 m in 0.10 ms, giving speed $0.60/0.00010 = 6000 \text{ m s}^{-1}$. The crack echo travels $6000 \times 0.000020 = 0.12 \text{ m}$ in total, so its depth is half of that: **0.06 m**.

Alternatively, both times are for return journeys at the same speed, so the depth ratio equals the time ratio: $0.30 \times (0.020/0.10) = 0.06 \text{ m}$.

Useful approach: Every echo time represents a journey to the reflector and back.

Question 25

A lamp has the following current–voltage readings:

Voltage across lamp	Current
2 V	0.30 A
4 V	0.50 A
6 V	0.60 A

The lamp is connected to a 12 V ideal supply through a $20\ \Omega$ resistor. Assuming the lamp current increases continuously with voltage throughout the operating range, which interval must contain the circuit current?

A Less than 0.20 A

B Between 0.20 A and 0.30 A

C Between 0.30 A and 0.50 A

D Between 0.50 A and 0.60 A

E Greater than 0.60 A

Worked solution

Answer C

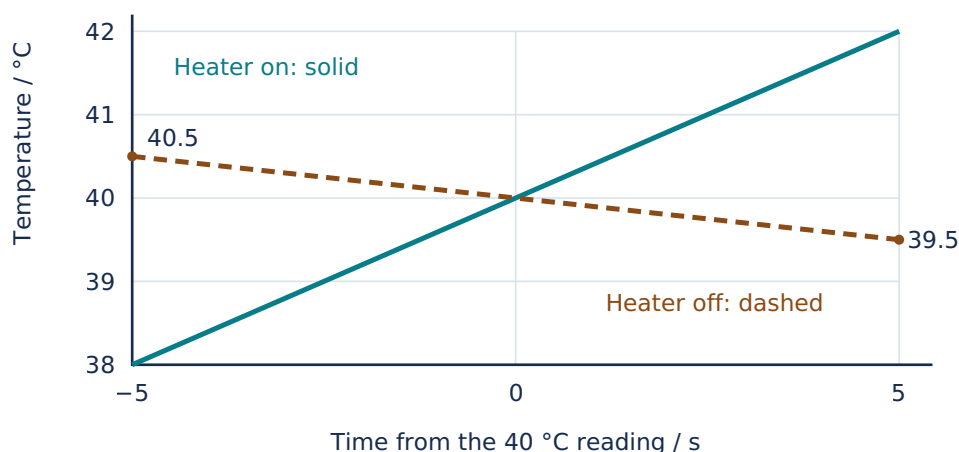
At 0.30 A, the resistor and lamp together would require $20 \times 0.30 + 2 = 8\text{ V}$, below 12 V, so current must be higher. At 0.50 A they would require $20 \times 0.50 + 4 = 14\text{ V}$, above 12 V, so current must be lower. It lies between 0.30 A and 0.50 A.

Useful approach: Test candidate currents in the total-voltage equation; do not replace the lamp with a fixed resistance.

Question 26

A metal block of mass 0.30 kg is heated by a 60 W heater. A second measurement is made while the same block cools with the heater off.

The straight lines below are tangents to the two temperature-time curves at 40°C . The rate of heat loss at this temperature is the same in both measurements. Neglect the heat capacities of the heater and thermometer, and assume the block has a uniform temperature.



What is the specific heat capacity of the metal?

A $200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

B $400 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

C $500 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

D $667 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

E $2000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

Worked solution

Answer B

The heating gradient is $(42 - 38)/10 = 0.40 \text{ }^\circ\text{C s}^{-1}$. The magnitude of the cooling gradient is $(40.5 - 39.5)/10 = 0.10 \text{ }^\circ\text{C s}^{-1}$.

Let H be the rate of heat loss at 40°C , and let c be the specific heat capacity. During cooling, $H = mc(0.10)$. With the heater on, $60 - H = mc(0.40)$.

Adding these energy-rate equations gives $60 = mc(0.50)$, so

$$c = \frac{60}{0.30 \times 0.50} = 400 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}.$$

Useful approach: At the same temperature, add the magnitudes of the heating and cooling slopes to correct for heat loss.

Question 27

A cart of mass m is released from rest and descends a frictionless track through a vertical height h . At the bottom it sticks to a stationary cart of mass $2m$. The combined carts then rise frictionlessly. What maximum vertical height do they reach above the bottom?

A $h/18$

B $h/9$

C $h/3$

D $h/2$

E $2h/3$

Worked solution

Answer B

Before the collision, $mgh = \frac{1}{2}mv^2$, so $v^2 = 2gh$. Momentum during the collision gives $mv = 3mu$, so $u = v/3$. After the collision, $\frac{1}{2}(3m)u^2 = (3m)gH$, hence $H = u^2/(2g) = h/9$.

Useful approach: Do not conserve kinetic energy through the sticking collision; use momentum only at that instant.

MODULE 03

Chemistry

QUESTIONS & WORKED SOLUTIONS

Physics + Chemistry pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

A dilute solution of hydrochloric acid dissociates completely. Two separate portions are diluted with water. The observations are shown below.

Dilution	Methyl orange	Bromothymol blue
Final volume is 10 times the original volume	red	yellow
Final volume is 100 times the original volume	yellow	yellow

Assume sharp indicator changes: methyl orange is red below pH 3.7 and yellow above pH 3.7; bromothymol blue is yellow below pH 6.5 and blue above pH 6.5. None of the solutions is exactly at a transition pH. Ignore the contribution of water to the hydrogen-ion concentration. For these solutions, $[H^+] = 10^{-n} \text{ mol dm}^{-3}$ corresponds to pH n .

Which could be the original hydrogen-ion concentration, in mol dm^{-3} ?

A 10^{-1}

B 10^{-2}

C 10^{-3}

D 10^{-4}

E 10^{-5}

Worked solution

Answer B

A tenfold dilution lowers the hydrogen-ion concentration by a factor of 10, so it raises pH by 1. A hundredfold dilution raises pH by 2.

If the original pH is p , the methyl-orange observations give $p + 1 < 3.7$ and $p + 2 > 3.7$. Therefore $1.7 < p < 2.7$.

Of the options, only $10^{-2} \text{ mol dm}^{-3}$ has a pH in this interval: pH 2. Its diluted solutions have pH 3 and 4, both below the bromothymol-blue transition. **Answer: B.**

Question 02

Atoms of elements X and Y have electron configurations 2,8,2 and 2,8,7 respectively. Z is the element immediately below Y in the same group of the Periodic Table.

Which row gives the formula of the ionic compound formed by X and Y, and the expected result when Y_2 is added to an aqueous solution of NaZ?

A XY; Y_2 displaces Z from Z^- ions.

B XY_2 ; no displacement occurs.

C X_2Y ; Y_2 displaces Z from Z^- ions.

D XY_2 ; Y_2 displaces Z from Z^- ions.

E X_2Y ; no displacement occurs.

Worked solution

Answer D

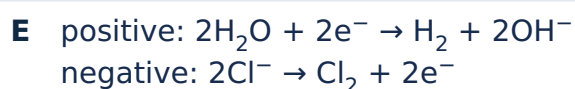
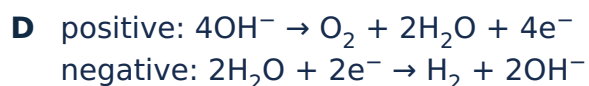
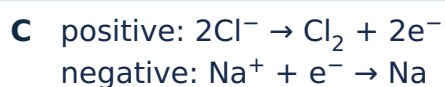
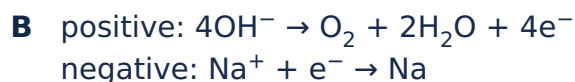
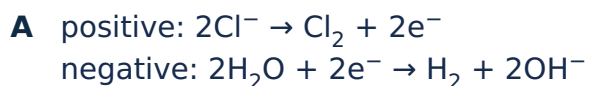
X loses its two outer electrons to form X^{2+} . Y gains one electron to form Y^- . Two Y^- ions balance each X^{2+} , so the formula is XY_2 .

Y is a Group 17 halogen. Halogen reactivity decreases down the group, so Y is more reactive than Z. The displacement is $Y_2 + 2Z^- \rightarrow 2Y^- + Z_2$. **Answer: D.**

Question 03

Concentrated aqueous sodium chloride is electrolysed using inert electrodes. The gas at the positive electrode bleaches damp litmus paper. The gas at the negative electrode gives a squeaky pop with a lighted splint.

Which row is correct?



Worked solution

Answer A

The positive electrode is the anode. Chloride ions lose electrons there, producing the chlorine identified by its bleaching action: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$.

The negative electrode is the cathode. In aqueous sodium chloride, water is reduced to hydrogen rather than sodium ions being reduced to sodium: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$. Hydrogen gives the squeaky pop. **Answer: A.**

Question 04

Two pure substances have the following properties.

Property	P	Q
Melting point / °C	-114	801
Boiling point / °C	-85	1465
Conducts when pure and liquid?	no	yes
Conducts when dissolved in water?	yes	yes

Which explanation is consistent with these observations?

- A** P and Q both consist of ions; the ions in pure liquid P cannot move.
- B** P contains delocalised electrons only in its gas phase; Q contains molecules.
- C** P consists of small molecules that form ions on reacting with water; Q contains ions that become mobile when molten or dissolved.
- D** P has a giant covalent structure that breaks into ions on melting; Q conducts through delocalised electrons.
- E** P consists of small molecules whose electrons become delocalised in water; Q consists of small molecules that become mobile on melting.

Worked solution

Answer C

P has very low melting and boiling points and does not conduct as a pure liquid, consistent with small covalent molecules. Its aqueous solution can conduct because those molecules react with water to form ions; aqueous hydrogen chloride is an example.

Q has a high melting point and conducts when molten or dissolved, as expected for an ionic substance with mobile ions. Conductivity of a solution alone does not show that the undissolved substance was ionic. **Answer: C.**

Question 05

A compound contains only carbon, hydrogen and oxygen. Complete combustion of 0.900 g of the compound produces 1.32 g of carbon dioxide and 0.540 g of water. Its relative molecular mass is 90.

Use $A_r(\text{C}) = 12$, $A_r(\text{H}) = 1$ and $A_r(\text{O}) = 16$. What is its molecular formula?

A CH_2O

B $\text{C}_2\text{H}_4\text{O}_2$

C $\text{C}_3\text{H}_8\text{O}_2$

D $\text{C}_6\text{H}_{12}\text{O}_6$

E $\text{C}_3\text{H}_6\text{O}_3$

Worked solution

Answer E

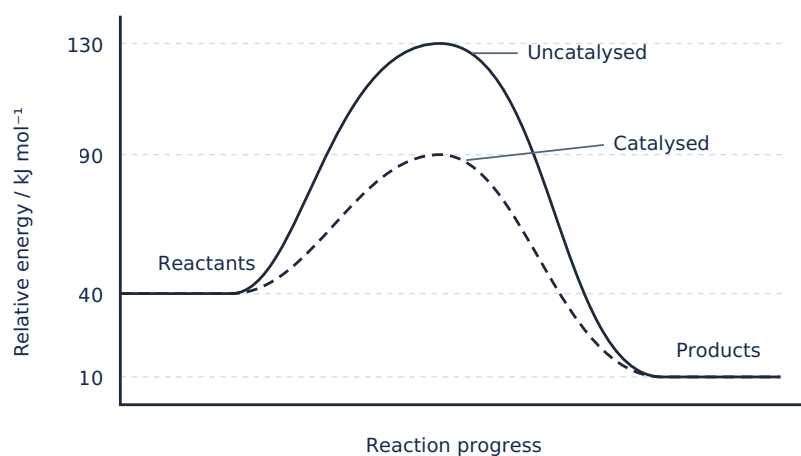
Carbon dioxide gives $n(\text{C}) = 1.32/44 = 0.0300$ mol, so the sample contains 0.360 g of carbon. Water gives $n(\text{H}) = 2 \times 0.540/18 = 0.0600$ mol, equivalent to 0.0600 g of hydrogen.

The oxygen in the original sample therefore has mass $0.900 - 0.360 - 0.0600 = 0.480$ g, or 0.0300 mol. The atom ratio C:H:O is $0.0300:0.0600:0.0300 = 1:2:1$.

The empirical formula CH_2O has formula mass 30. Since $90/30 = 3$, the molecular formula is $\text{C}_3\text{H}_6\text{O}_3$. **Answer: E.**

Question 06

The diagram shows two pathways between the same reactants and products.



Which row gives the activation energy of the **catalysed reverse reaction** and the energy change, ΔH , of the **forward reaction**, in kJ mol^{-1} ?

A Activation energy 50; $\Delta H = -30$

B Activation energy 80; $\Delta H = -30$

C Activation energy 80; $\Delta H = +30$

D Activation energy 120; $\Delta H = -30$

E Activation energy 90; $\Delta H = +30$

Worked solution

Answer B

The reverse reaction starts at the product level, 10 kJ mol^{-1} . The catalysed pathway reaches 90 kJ mol^{-1} , so its reverse activation energy is $90 - 10 = 80 \text{ kJ mol}^{-1}$.

For the forward reaction, $\Delta H = \text{energy of products} - \text{energy of reactants} = 10 - 40 = -30 \text{ kJ mol}^{-1}$. The catalyst changes the pathway, not the endpoint energies. **Answer: B.**

Question 07

A sample of element X contains only the isotopes ^{54}X , ^{56}X and ^{58}X . There are twice as many ^{58}X atoms as ^{54}X atoms. The relative atomic mass of X in this sample is 56.2.

What percentage of the atoms are ^{56}X ? Treat each isotope mass as its mass number.

A 10%

B 20%

C 60%

D 70%

E 80%

Worked solution

Answer D

Let the fraction of ^{54}X be x . The fractions of ^{58}X and ^{56}X are then $2x$ and $1 - 3x$.

The weighted mean is $54x + 56(1 - 3x) + 58(2x) = 56 + 2x = 56.2$. Hence $x = 0.10$.

The fraction of ^{56}X is $1 - 3(0.10) = 0.70$, or 70%. **Answer: D.**

Question 08

Copper(II) ions react with iodide ions to form solid copper(I) iodide and iodine. Use the processes below.



A total of 0.0150 mol of Cu^{2+} reacts, and all of this copper ends in CuI. Ignore subsequent reactions of iodine. Which row is correct?

A I^{-} consumed: 0.0150 mol; I_2 formed: 0.00750 mol; reducing agent: I^{-} .

B I^{-} consumed: 0.0300 mol; I_2 formed: 0.0150 mol; reducing agent: I^{-} .

C I^{-} consumed: 0.0300 mol; I_2 formed: 0.00750 mol; reducing agent: I^{-} .

D I^{-} consumed: 0.0300 mol; I_2 formed: 0.00750 mol; reducing agent: Cu^{2+} .

E I^{-} consumed: 0.0450 mol; I_2 formed: 0.0150 mol; reducing agent: Cu^{2+} .

Worked solution

Answer C

Each Cu^{2+} gains one electron, so 0.0150 mol of electrons is needed. Oxidation of 0.0150 mol of I^{-} supplies these electrons and makes 0.00750 mol of I_2 .

A further 0.0150 mol of I^{-} combines with the Cu^{+} to make CuI. The total iodide consumption is therefore 0.0300 mol. Iodide loses electrons and is the reducing agent.

The overall equation is $2\text{Cu}^{2+} + 4\text{I}^{-} \rightarrow 2\text{CuI(s)} + \text{I}_2$. **Answer: C.**

Question 09

An addition polymer is made using only ethene, C_2H_4 , and chloroethene, $\text{C}_2\text{H}_3\text{Cl}$. No atoms are lost when either monomer joins the chain.

A representative section of the polymer contains 120 carbon atoms and 20 chlorine atoms. Ignore the ends of the polymer chain.

What fraction of the incorporated monomer molecules were chloroethene, and how many hydrogen atoms are in this section?

A Fraction $1/6$; 220 hydrogen atoms

B Fraction $1/3$; 240 hydrogen atoms

C Fraction $1/2$; 200 hydrogen atoms

D Fraction $2/3$; 200 hydrogen atoms

E Fraction $1/3$; 220 hydrogen atoms

Worked solution

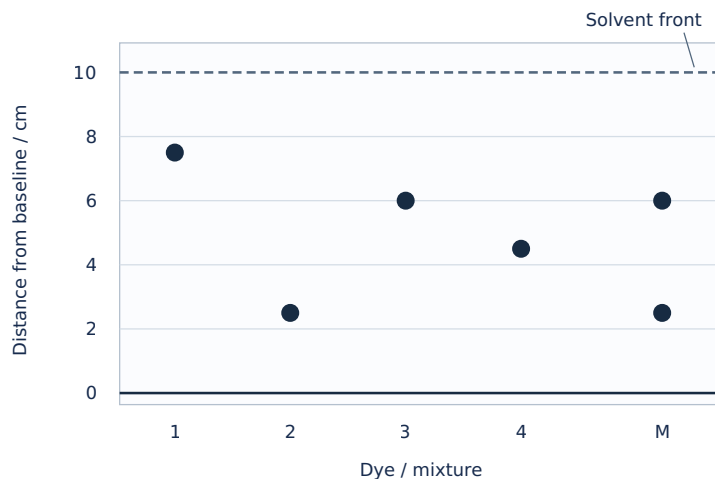
Answer E

Both monomers supply two carbon atoms, so 120 carbon atoms represent 60 monomer molecules. Only chloroethene supplies chlorine; 20 chlorine atoms therefore represent 20 chloroethene molecules.

The chloroethene fraction is $20/60 = 1/3$. The section contains 40 ethene units and 20 chloroethene units, giving $40 \times 4 + 20 \times 3 = 220$ hydrogen atoms. **Answer: E.**

Question 10

Four pure dyes and mixture M are run on the same chromatogram. M contains only one or more of these four dyes, and every component present gives a visible spot under these conditions.



Consider the statements.

1. The R_f value of dye 3 is 0.60.
2. The filter paper is the mobile phase.
3. M contains dyes 2 and 3.

Which statements are correct?

A 1 only

B 2 only

C 3 only

D 1 and 3 only

E 1, 2 and 3

Worked solution

Answer D

Dye 3 has travelled 6.0 cm and the solvent front 10.0 cm, so $R_f = 6.0/10.0 = 0.60$. Statement 1 is true.

The solvent is the mobile phase; the paper is stationary, so statement 2 is false. The spots in M match dyes 2 and 3. The restriction to these four dyes, together with the stated visibility of all components, makes this identification justified. Statement 3 is true. **Answer: D.**

Question 11

A dry mixture of helium and carbon dioxide has mass 4.00 g and occupies 2.40 dm³ at room temperature and pressure.

Treat both gases as ideal. Use a molar gas volume of 24.0 dm³ mol⁻¹, $A_r(\text{He}) = 4$ and $M_r(\text{CO}_2) = 44$.

What volume would the carbon dioxide alone occupy at the same temperature and pressure?

A 0.240 dm³

B 1.20 dm³

C 2.16 dm³

D 1.92 dm³

E 2.40 dm³

Worked solution

Answer C

The total amount of gas is $2.40/24.0 = 0.100$ mol. Let the amount of CO₂ be x mol, leaving $(0.100 - x)$ mol of helium.

The mass equation is $44x + 4(0.100 - x) = 4.00$. Thus $40x = 3.60$ and $x = 0.0900$ mol.

The carbon dioxide volume is $0.0900 \times 24.0 = 2.16$ dm³. **Answer: C.**

Question 12

A 25.0 cm^3 sample of sodium hydroxide solution is mixed with 40.0 cm^3 of $0.150 \text{ mol dm}^{-3}$ hydrochloric acid. The acid is in excess.

The remaining acid requires 10.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ sodium hydroxide for complete neutralisation.

What was the concentration of the original sodium hydroxide solution?

A $0.0800 \text{ mol dm}^{-3}$

B $0.160 \text{ mol dm}^{-3}$

C $0.240 \text{ mol dm}^{-3}$

D $0.320 \text{ mol dm}^{-3}$

E $0.400 \text{ mol dm}^{-3}$

Worked solution

Answer B

The initial amount of HCl is $0.0400 \times 0.150 = 0.00600 \text{ mol}$. The later titration uses $0.0100 \times 0.200 = 0.00200 \text{ mol}$ of NaOH, so 0.00200 mol of HCl remained.

The original sample neutralised $0.00600 - 0.00200 = 0.00400 \text{ mol}$ of HCl. Since HCl and NaOH react 1:1, it contained 0.00400 mol of NaOH.

Its concentration was $0.00400/0.0250 = 0.160 \text{ mol dm}^{-3}$. **Answer: B.**

Question 13

Chlorine reacts with hot concentrated sodium hydroxide. All the chlorine atoms end in chloride ions, Cl^- , or chlorate ions, ClO_3^- . No other element changes oxidation state.

If 0.120 mol of Cl_2 reacts, how many moles of Cl^- form?

A 0.0200 mol

B 0.0400 mol

C 0.100 mol

D 0.120 mol

E 0.200 mol

Worked solution

Answer E

Chlorine starts with oxidation state 0. It changes to -1 in Cl^- , gaining one electron per atom. In ClO_3^- , chlorine is $+5$ because three oxygen atoms contribute -6 and the ion has charge -1 .

Each chlorine atom oxidised to $+5$ loses five electrons, so five chlorine atoms must be reduced to -1 . The chloride:chlorate ratio is 5:1.

The 0.120 mol of Cl_2 contains 0.240 mol of chlorine atoms. Five-sixths enter chloride, giving $(5/6) \times 0.240 = 0.200$ mol. **Answer: E.**

Question 14

A carbonate reacts with dilute acid at constant temperature. Only carbon dioxide escapes from the flask. The cumulative decrease in the balance reading is recorded.

Time / s	0	20	40	60	80
Mass lost / mg	0	16	28	34	36

Which row correctly gives the average mass-loss rates over 0–40 s and 40–80 s, together with a possible explanation for the later change in rate?

- A** 0.70 and 0.20 mg s⁻¹; acid depletion reduces the frequency of collisions between acid particles and the carbonate.
- B** 0.70 and 0.20 mg s⁻¹; acid depletion lowers the fraction of acid particles with energy above the activation energy.
- C** 0.70 and 0.20 mg s⁻¹; acid depletion lowers the mean kinetic energy of the remaining acid particles.
- D** 1.40 and 0.40 mg s⁻¹; acid depletion reduces the frequency of collisions between acid particles and the carbonate.
- E** 0.20 and 0.70 mg s⁻¹; acid depletion reduces the frequency of collisions between acid particles and the carbonate.

Worked solution

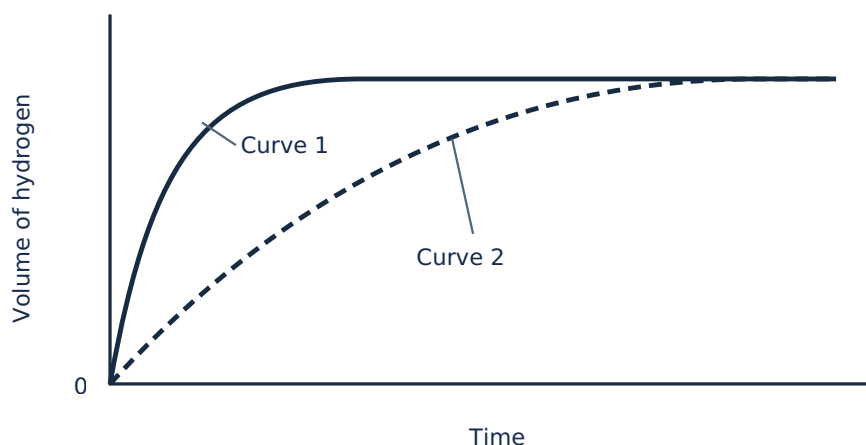
Answer A

The first average rate is $28/40 = 0.70 \text{ mg s}^{-1}$. The later interval loses an additional $36 - 28 = 8 \text{ mg}$ in 40 s, giving $8/40 = 0.20 \text{ mg s}^{-1}$.

As acid is consumed, fewer acid particles per unit volume can reach the carbonate, reducing collision frequency. At constant temperature, acid depletion does not by itself lower the mean kinetic energy or the fraction of particles above a fixed activation energy. **Answer: A.**

Question 15

Curve 2 shows hydrogen collected when magnesium chips react with excess dilute hydrochloric acid. All magnesium eventually reacts.



Each of the following changes is considered separately.

1. Use the same mass of magnesium as a powder.
2. Use twice the mass of magnesium chips of the same size.
3. Use the original magnesium chips but double the acid concentration, keeping the acid volume the same.

Acid remains in excess in every experiment. Temperature is kept constant, and all gas volumes are measured at the same temperature and pressure. Which changes could produce curve 1?

A 1 only

B 2 only

C 3 only

D 1 and 2 only

E 1 and 3 only

F 2 and 3 only

Worked solution

Answer E

Curve 1 has a higher initial rate but the same final hydrogen volume. With acid in excess, the amount of magnesium fixes the final amount of hydrogen.

Change 1 increases magnesium surface area without changing its amount, so it can increase rate while preserving the final volume. Change 3 increases acid concentration and collision frequency, again without changing the limiting amount of magnesium.

Change 2 doubles the amount of magnesium and would double the final hydrogen amount. It cannot give the shared plateau. Thus changes 1 and 3 are suitable.

Answer: E.

Question 16

The solubility of a non-volatile salt X is given below.

Temperature / °C	Solubility / g of X per 100 g of water
60	60
20	20

A 200 g sample of saturated solution at 60 °C is cooled to 20 °C. During cooling, 10.0 g of water evaporates. The mixture is allowed to reach equilibrium, and the resulting crystals contain no water.

What mass of X crystallises?

A 32.0 g

B 40.0 g

C 52.0 g

D 50.0 g

E 55.0 g

Worked solution

Answer C

At 60 °C, 160 g of saturated solution contains 100 g of water and 60 g of X. The 200 g sample therefore contains 125 g of water and 75 g of X.

After evaporation, 115 g of water remains. At 20 °C this can dissolve $(20/100) \times 115 = 23.0$ g of X.

The crystallised mass is $75.0 - 23.0 = 52.0$ g. The 200 g initial mass is the solution mass, not the water mass. **Answer: C.**

Question 17

Use the following gas-phase reactions and energy changes.



The mean bond energies are $\text{H-H} = 436$, $\text{Cl-Cl} = 242$ and $\text{C-H} = 414 \text{ kJ mol}^{-1}$. Use the same mean energy for every C-H bond in this calculation.

What value of the C-Cl bond energy follows from these data?

A 225 kJ mol⁻¹

B 335 kJ mol⁻¹

C 431 kJ mol⁻¹

D 445 kJ mol⁻¹

E 519 kJ mol⁻¹

Worked solution

Answer B

Using $\Delta H = \text{energy required to break bonds} - \text{energy released on forming bonds}$, the first reaction gives $-184 = 436 + 242 - 2D(\text{H-Cl})$. Hence $D(\text{H-Cl}) = 431 \text{ kJ mol}^{-1}$.

In the second reaction, three C-H bonds occur on both sides and cancel under the stated mean-bond approximation. The net changes break one C-H and one Cl-Cl bond and form one C-Cl and one H-Cl bond.

Therefore $-110 = 414 + 242 - D(\text{C-Cl}) - 431$, giving $D(\text{C-Cl}) = 335 \text{ kJ mol}^{-1}$.

Answer: B.

Question 18

A 100 g mixture of hexane and octane contains 40.0% hexane by mass. During fractional distillation, the first 20.0 g of distillate collected contains 75.0% hexane by mass. Assume all material is either in the flask or in the collected distillate.

Hexane boils at 69 °C and octane at 126 °C. Which row gives the percentage of hexane by mass remaining in the flask and identifies the more volatile component?

A 25.0%; hexane

B 31.25%; octane

C 50.0%; hexane

D 31.25%; hexane

E 75.0%; octane

Worked solution

Answer D

The original mixture contains 40.0 g of hexane. The distillate removes $0.750 \times 20.0 = 15.0$ g of hexane, leaving 25.0 g.

The flask now contains $100 - 20.0 = 80.0$ g of mixture, so its hexane content is $(25.0/80.0) \times 100\% = 31.25\%$.

Hexane has the lower boiling point and is the more volatile component, consistent with its enrichment in the first distillate. The stated boiling points alone would not determine the exact temperature of the mixture in the flask. **Answer: D.**

Question 19

An 80.0 g sample of anhydrous CuSO_4 , $M_r = 160$, is dissolved to make 500 cm^3 of solution. The solution is electrolysed using inert electrodes, and 8.0 g of copper is deposited at the cathode.

Assume that no copper enters the solution at the anode, that the solution volume remains 500 cm^3 , and that the only process removing Cu^{2+} is its reduction to copper metal. Use $A_r(\text{Cu}) = 64$.

What is the final concentration of Cu^{2+} ions?

A 0.25 mol dm^{-3}

B 0.50 mol dm^{-3}

C 1.00 mol dm^{-3}

D 1.25 mol dm^{-3}

E 0.75 mol dm^{-3}

Worked solution

Answer E

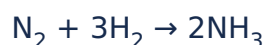
Initially, $n(\text{Cu}^{2+}) = n(\text{CuSO}_4) = 80.0/160 = 0.500 \text{ mol}$. The copper deposit is $8.0/64 = 0.125 \text{ mol}$, so 0.125 mol of Cu^{2+} has been removed.

There are $0.500 - 0.125 = 0.375 \text{ mol}$ of Cu^{2+} left in 0.500 dm^3 . Its concentration is $0.375/0.500 = 0.750 \text{ mol dm}^{-3}$.

This calculation tracks copper ions. It does not mean that sulfate ions have been removed together with the deposited copper. **Answer: E.**

Question 20

A sample of nitrogen occupies 4.80 dm^3 at room temperature and pressure. It reacts with excess hydrogen according to:



In this stage, 75.0% of the nitrogen reacts. All the ammonia produced is then converted to ammonium nitrate:



Only 80.0% of the ammonium nitrate formed is recovered as a product. Use a molar gas volume of $24.0 \text{ dm}^3 \text{ mol}^{-1}$ at room temperature and pressure and $M_r(\text{NH}_4\text{NO}_3) = 80$.

What mass is recovered?

A 9.60 g

B 14.4 g

C 19.2 g

D 24.0 g

E 25.6 g

Worked solution

Answer C

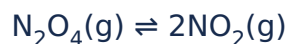
The initial amount of nitrogen is $4.80/24.0 = 0.200 \text{ mol}$. The amount that reacts is $0.750 \times 0.200 = 0.150 \text{ mol}$.

Each mole of nitrogen produces two moles of ammonia, so 0.300 mol of NH_3 forms. This makes 0.300 mol of ammonium nitrate, with mass $0.300 \times 80 = 24.0 \text{ g}$.

The recovered mass is $0.800 \times 24.0 = 19.2 \text{ g}$. The first percentage applies to nitrogen conversion and the second to recovery of the salt that actually forms. **Answer: C.**

Question 21

A sealed vessel contains an equilibrium mixture of the ideal gases N_2O_4 and NO_2 :



The vessel volume is suddenly halved at constant temperature. Consider the instant after compression, before reaction changes the amounts, and then the new equilibrium.

Which row correctly describes the NO_2 concentration at that instant, and the subsequent change in the **number of NO_2 molecules** as equilibrium is re-established?

- A** Concentration doubles; number of molecules subsequently decreases.
- B** Concentration doubles; number of molecules subsequently increases.
- C** Concentration is unchanged; number of molecules subsequently decreases.
- D** Concentration halves; number of molecules subsequently increases.
- E** Concentration doubles; number of molecules subsequently stays unchanged.

Worked solution

Answer A

Immediately after compression, the amount of NO_2 has not yet changed but the volume is half as large. Its concentration therefore doubles.

The increased pressure favours the side with fewer gas molecules. This is the left side, with one N_2O_4 for every two NO_2 on the right. The subsequent net reaction consumes NO_2 , so the number of NO_2 molecules decreases.

The question distinguishes a concentration change caused immediately by compression from a later change in chemical amount. **Answer: A.**

Question 22

A 20.0 cm^3 mixture containing only methane, CH_4 , and ethane, C_2H_6 , is burned completely in excess oxygen. It produces 30.0 cm^3 of carbon dioxide.

All gas volumes are compared at the same temperature and pressure. What volume of oxygen is consumed?

A 40.0 cm^3

B 45.0 cm^3

C 50.0 cm^3

D 55.0 cm^3

E 60.0 cm^3

Worked solution

Answer D

Let methane and ethane volumes be x and y . The total volume gives $x + y = 20.0$. Methane produces one CO_2 per molecule and ethane two, so $x + 2y = 30.0$. Thus both gases occupy 10.0 cm^3 .

The combustion equations are $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ and $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$.

The oxygen volume is $2(10.0) + (7/2)(10.0) = 55.0 \text{ cm}^3$. **Answer: D.**

Question 23

Metal P displaces Q from an aqueous salt solution, and metal Q displaces R. Carbon reduces an oxide of R but cannot reduce an oxide of Q under appropriate extraction conditions.

Use the usual reactivity-series model: an element displaces a less reactive element from its compound. What is the order from most to least reactive?

A $P > C > Q > R$

B $P > Q > C > R$

C $C > P > Q > R$

D $P > Q > R > C$

E $Q > P > C > R$

Worked solution

Answer B

The metal displacements give $P > Q > R$. Carbon reducing R oxide places C above R, while its inability to reduce Q oxide places C below Q within the stated reactivity-series model.

Combining these relations gives $P > Q > C > R$. Each neighbouring relation is needed to distinguish the alternatives. **Answer: B.**

Question 24

An impure sample contains calcium carbonate and material that does not react with acid. Excess acid produces 0.600 dm^3 of carbon dioxide at room temperature and pressure. All this gas comes from the calcium carbonate and is collected.

The sample is 75.0% calcium carbonate by mass. Use a molar gas volume of $24.0 \text{ dm}^3 \text{ mol}^{-1}$ and $M_r(\text{CaCO}_3) = 100$.

What was the sample mass, to the nearest 0.01 g?

A 2.00 g

B 2.67 g

C 3.33 g

D 4.00 g

E 5.00 g

Worked solution

Answer C

$\text{CaCO}_3 + 2\text{H}^+ \rightarrow \text{Ca}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$ gives one mole of CO_2 per mole of calcium carbonate.

The gas amount is $0.600/24.0 = 0.0250 \text{ mol}$, corresponding to $0.0250 \times 100 = 2.50 \text{ g}$ of CaCO_3 .

This is 75.0% of the sample mass, so the sample mass is $2.50/0.750 = 3.333\dots \text{ g}$, or 3.33 g. **Answer: C.**

Question 25

An aqueous solution contains one dissolved salt. Separate portions are tested.

- Sodium hydroxide gives a blue precipitate that is insoluble in excess sodium hydroxide.
- After acidification with dilute nitric acid, silver nitrate gives a cream precipitate.

Which salt is most consistent with both observations?

A CuCl

B CuCl₂

C FeBr₂

D CuBr₂

E NaBr

Worked solution

Answer D

The blue precipitate is consistent with copper(II) hydroxide, identifying Cu²⁺ ions. The cream silver-halide precipitate is silver bromide, identifying Br⁻ ions.

One Cu²⁺ requires two Br⁻ ions for charge balance, giving CuBr₂. Dilute nitric acid is specified so that the acid itself does not introduce chloride ions. **Answer: D.**

Question 26

A catalytic converter promotes the reaction:



When 0.100 mol of NO reacts, which row correctly identifies the reducing agent and the amount of electrons transferred from that agent to the species being reduced?

A CO; 0.100 mol of electrons

B NO; 0.100 mol of electrons

C NO; 0.200 mol of electrons

D CO; 0.400 mol of electrons

E CO; 0.200 mol of electrons

Worked solution

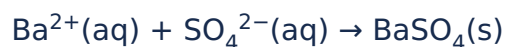
Answer E

Carbon changes from +2 in CO to +4 in CO₂. It loses two electrons per carbon atom, so CO is oxidised and acts as the reducing agent.

Nitrogen changes from +2 in NO to 0 in N₂, gaining two electrons per nitrogen atom. Therefore 0.100 mol of NO gains $2 \times 0.100 = 0.200$ mol of electrons. The same electron amount is supplied by CO. **Answer: E.**

Question 27

A 20.0 cm^3 sample of $0.100 \text{ mol dm}^{-3} \text{ BaCl}_2$ is mixed with 30.0 cm^3 of $0.100 \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4$. Barium sulfate precipitates:



Assume precipitation is complete and use $M_r(\text{BaSO}_4) = 233$. What mass of precipitate forms?

A 0.233 g

B 0.466 g

C 0.699 g

D 1.17 g

E 2.33 g

Worked solution

Answer B

The solutions contain $n(\text{Ba}^{2+}) = 0.0200 \times 0.100 = 0.00200 \text{ mol}$ and $n(\text{SO}_4^{2-}) = 0.0300 \times 0.100 = 0.00300 \text{ mol}$.

The precipitation ratio is 1:1. Barium ions are limiting, so 0.00200 mol of BaSO_4 forms. Its mass is $0.00200 \times 233 = 0.466 \text{ g}$. **Answer: B.**